

The empirical recurrence for OEIS A263814: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A263814 records “Empirical recurrence of order 82”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 1024$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 82, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 82 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A263814 is “Number of $(n+1)X(4+1)$ arrays of permutations of $0..n*5+4$ with each element having index change $(+,-,+,-)$ 0,0 0,1 0,2 or 1,0.”. It has offset 1 and begins

3682, 333429, 31916832, 3019386508, 286407546797, 27150876030959, 2574222711408930, 244058306583267424, .

The entry states:

Empirical recurrence of order 82 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 12:14:41, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1024$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 1024$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2048$ exact terms of the model returns order 82 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{82} c_i a(n-i),$$

c_1	92	c_{22}	−85150944102574240893	c_{43}	232148706450636115381782	c_{64}	76910
c_2	1106	c_{23}	−23844071016899915554	c_{44}	1024955672047994016905256	c_{65}	82258
c_3	−80332	c_{24}	607737787643216936160	c_{45}	−161295514588558010676482	c_{66}	−4758
c_4	−182305	c_{25}	46432387831196704574	c_{46}	−675006730548125204194644	c_{67}	−605
c_5	24596599	c_{26}	−3411881962936698654120	c_{47}	87160908233761470813738	c_{68}	2121
c_6	−41519541	c_{27}	204404892241397043402	c_{48}	360319362607726420494224	c_{69}	3134
c_7	−3455047852	c_{28}	15213157854343505162144	c_{49}	−36028659945193893315262	c_{70}	−64
c_8	12910590247	c_{29}	−2259580702204998209078	c_{50}	−155640527873477733391780	c_{71}	−11
c_9	253513305753	c_{30}	−54263167554541445266756	c_{51}	10995151765442482129870	c_{72}	123
c_{10}	−1250218917933	c_{31}	10995151765442482129870	c_{52}	54263167554541445266756	c_{73}	25
c_{11}	−11074591724091	c_{32}	155640527873477733391780	c_{53}	−2259580702204998209078	c_{74}	−1
c_{12}	64829352268879	c_{33}	−36028659945193893315262	c_{54}	−15213157854343505162144	c_{75}	−
c_{13}	313445173200737	c_{34}	−360319362607726420494224	c_{55}	204404892241397043402	c_{76}	−
c_{14}	−2121425294740601	c_{35}	87160908233761470813738	c_{56}	3411881962936698654120	c_{77}	−
c_{15}	−6054818134195786	c_{36}	675006730548125204194644	c_{57}	46432387831196704574	c_{78}	−
c_{16}	47582319688416311	c_{37}	−161295514588558010676482	c_{58}	−607737787643216936160	c_{79}	−
c_{17}	82258910420552655	c_{38}	−1024955672047994016905256	c_{59}	−23844071016899915554	c_{80}	−
c_{18}	−769101483534795177	c_{39}	232148706450636115381782	c_{60}	85150944102574240893	c_{81}	−
c_{19}	−794888427476631342	c_{40}	1262714518253483836677348	c_{61}	5389876464244224814	c_{82}	−
c_{20}	9263924863110530838	c_{41}	−261905351589489236062458	c_{62}	−9263924863110530838		
c_{21}	5389876464244224814	c_{42}	−1262714518253483836677348	c_{63}	−794888427476631342		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 82, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $82 + 4$ terms removed returns order 82 as well, so $\deg q_{\min} = 82 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $82 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 82 that this sequence satisfies — and that family turns out to have exactly one member.

References

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